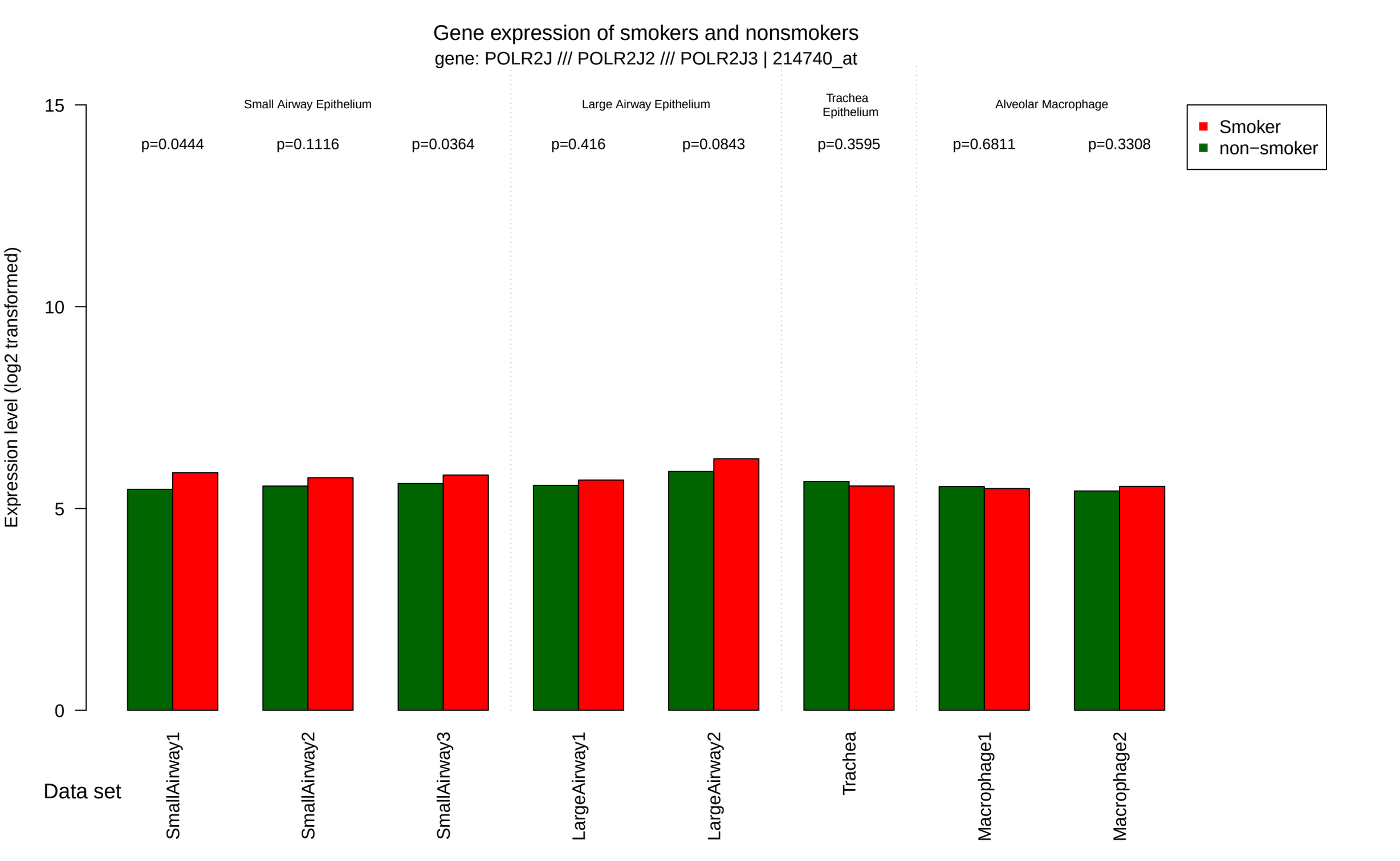
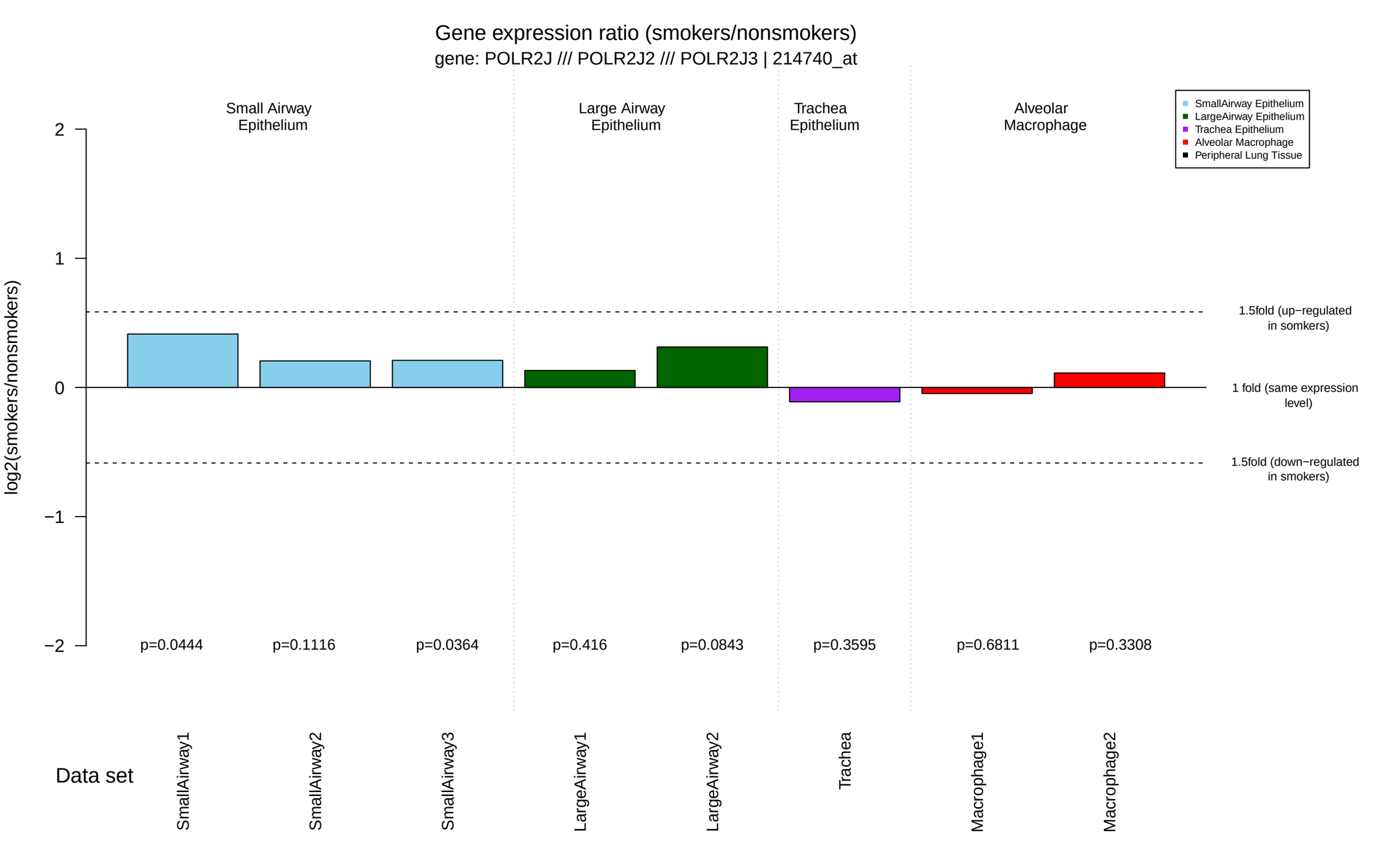
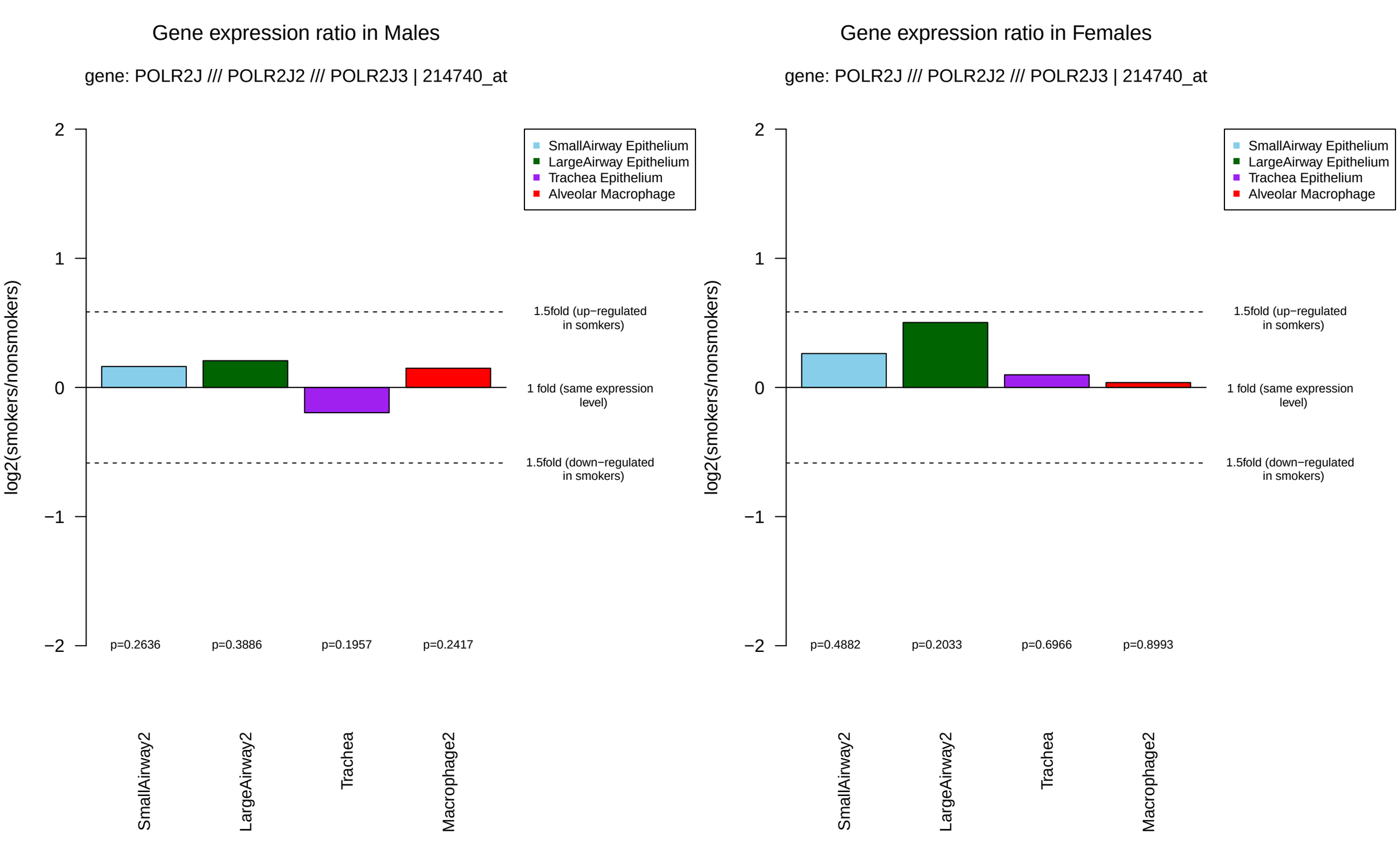
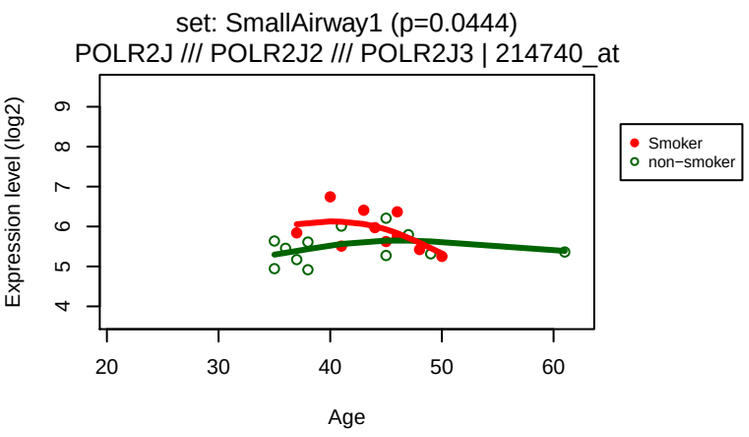
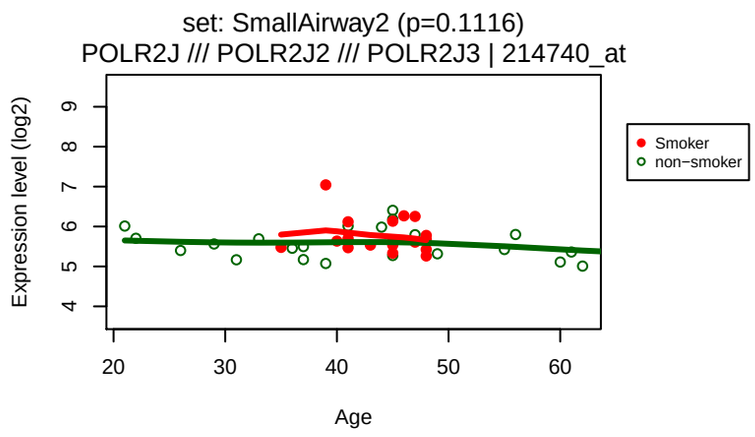
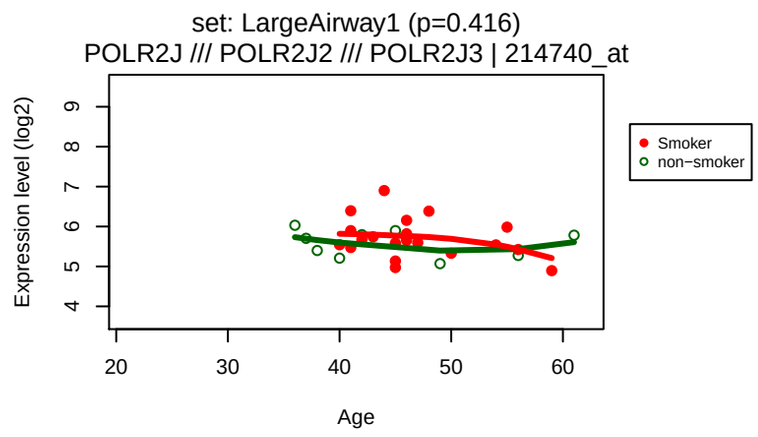

Gene expression of smokers and nonsmokers
gene: POLR2J /// POLR2J2 /// POLR2J3 | 214740_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0444
p=0.1116
p=0.0364
p=0.416
p=0.0843
p=0.3595
p=0.6811
p=0.3308
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

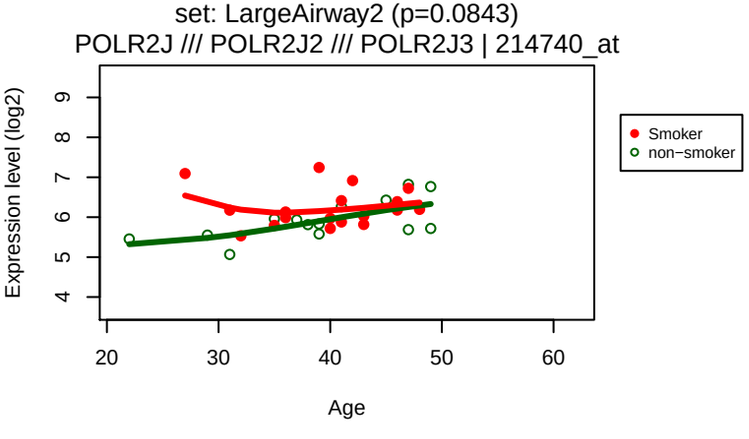

Gene expression ratio (smokers/nonsmokers)
gene: POLR2J /// POLR2J2 /// POLR2J3 | 214740_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0444
p=0.1116
p=0.0364
p=0.416
p=0.0843
p=0.3595
p=0.6811
p=0.3308
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

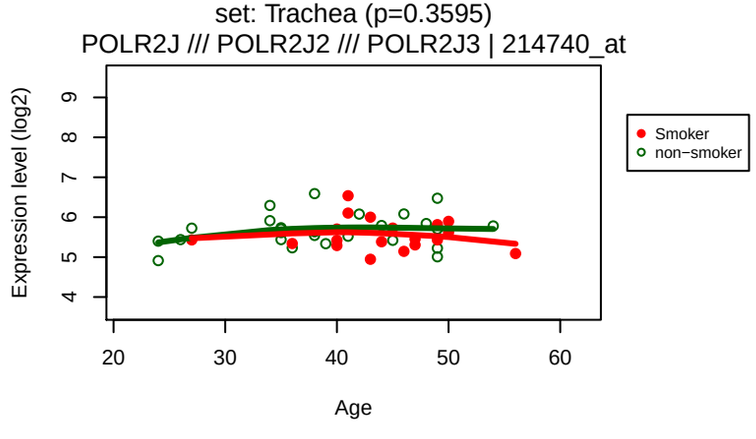

Gene expression ratio in Males
gene: POLR2J /// POLR2J2 /// POLR2J3 | 214740_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2636
p=0.3886
p=0.1957
p=0.2417
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLR2J /// POLR2J2 /// POLR2J3 | 214740_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4882
p=0.2033
p=0.6966
p=0.8993
SmallAirway2
LargeAirway2
Trachea
Macrophage2

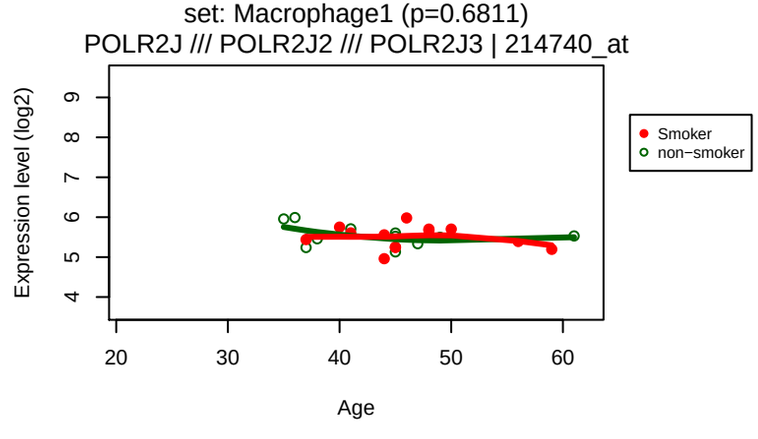

set: SmallAirway1 (p=0.0444)
POLR2J /// POLR2J2 /// POLR2J3 | 214740_at
Expression level (log2)
Age
Smoker
non−smoker

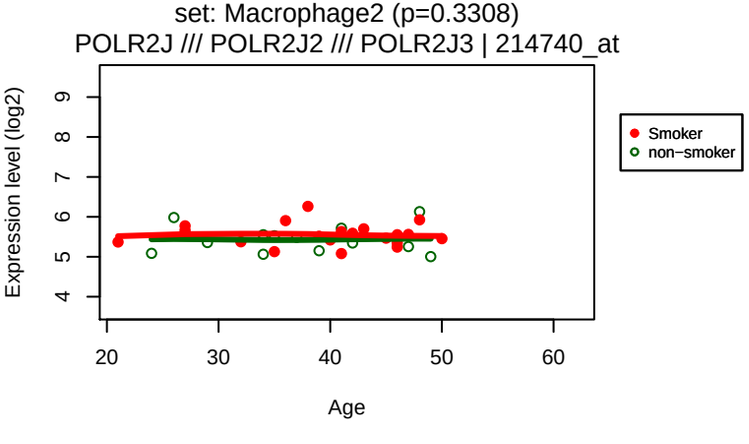

set: SmallAirway2 (p=0.1116)
POLR2J /// POLR2J2 /// POLR2J3 | 214740_at
Expression level (log2)
Age
Smoker
non−smoker

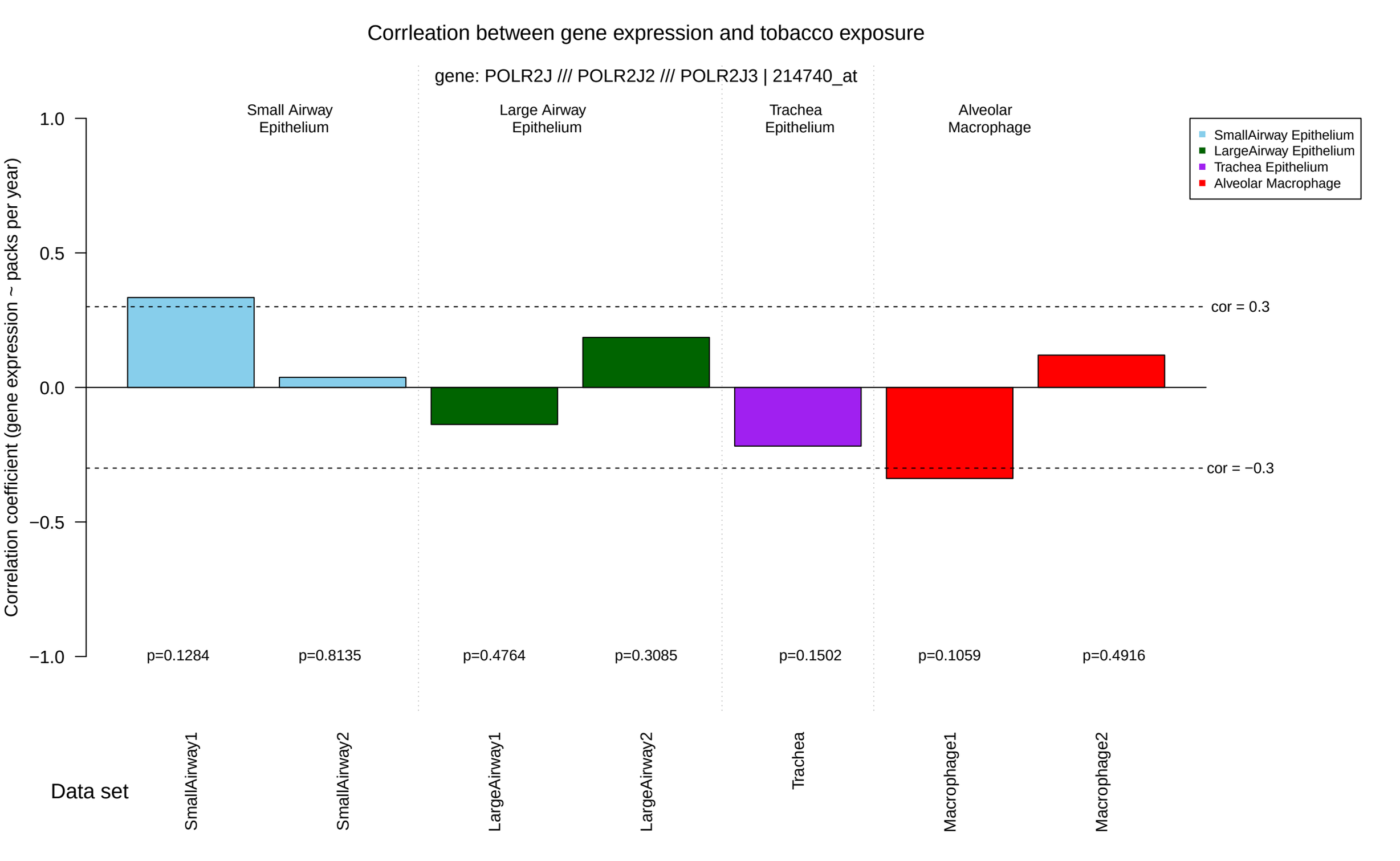

set: LargeAirway1 (p=0.416)
POLR2J /// POLR2J2 /// POLR2J3 | 214740_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0843)
POLR2J /// POLR2J2 /// POLR2J3 | 214740_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3595)
POLR2J /// POLR2J2 /// POLR2J3 | 214740_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6811)
POLR2J /// POLR2J2 /// POLR2J3 | 214740_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3308)
POLR2J /// POLR2J2 /// POLR2J3 | 214740_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLR2J /// POLR2J2 /// POLR2J3 | 214740_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1284
p=0.8135
p=0.4764
p=0.3085
p=0.1502
p=0.1059
p=0.4916
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2